
DATABASE DESIGN REPORT: FAST FOOD ORDERING SYSTEM

HIT 137 ASSESSMENT 3

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Part A: Business

Description and Rules

1.1. Business Situation: "Burger Haven" Ordering System

Burger Haven is an independent, medium-sized fast-food restaurant interested in replacing its current, manual and paper-based ticketing system with a digital database to handle its food ordering process. The proposed database will be used to track the customers, the employees who process the orders, the menu that is available and the specifics of every transaction.

The many-to-many relationship between orders and menu items may be a unique and potentially challenging element of this proposed application. Since one order may have multiple items, and a single item may be ordered in many different orders, the database must have an associative entity (Order Details) to accurately monitor the number of specific items in each order without causing a data redundancy (Coronel & Morris, 2018).

1.2. Business Rules

The following business rule and logical assumptions dictate the database design:

- **Customer and Orders:** A customer may have zero, one or multiple orders over time. Every order is ordered by only one customer.
- **Employees and Orders:** Each shift, an employee is allowed to process zero, one, or many orders. Only one employee processes each order.
- **Orders and Menu Items:** An order should have at least one menu item and can have many. The same menu item may be on none, one or numerous orders.
- **Order Details:** To resolve the many-to-many relationship between Orders and Menu Items, an entity called an Order Line or Order Detail will be used to record the exact quantity of each item ordered in a particular order.

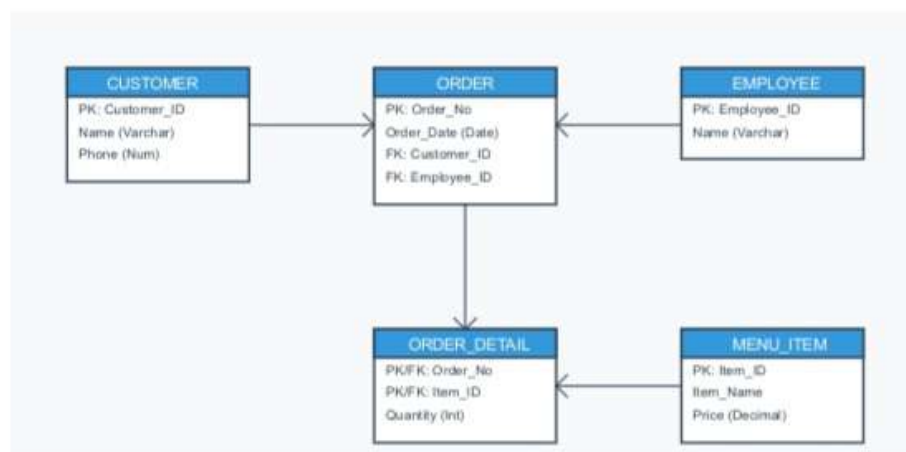


Figure 1.ER DIAGRAM

Part B: Map the ERD to EER Notation and Relations

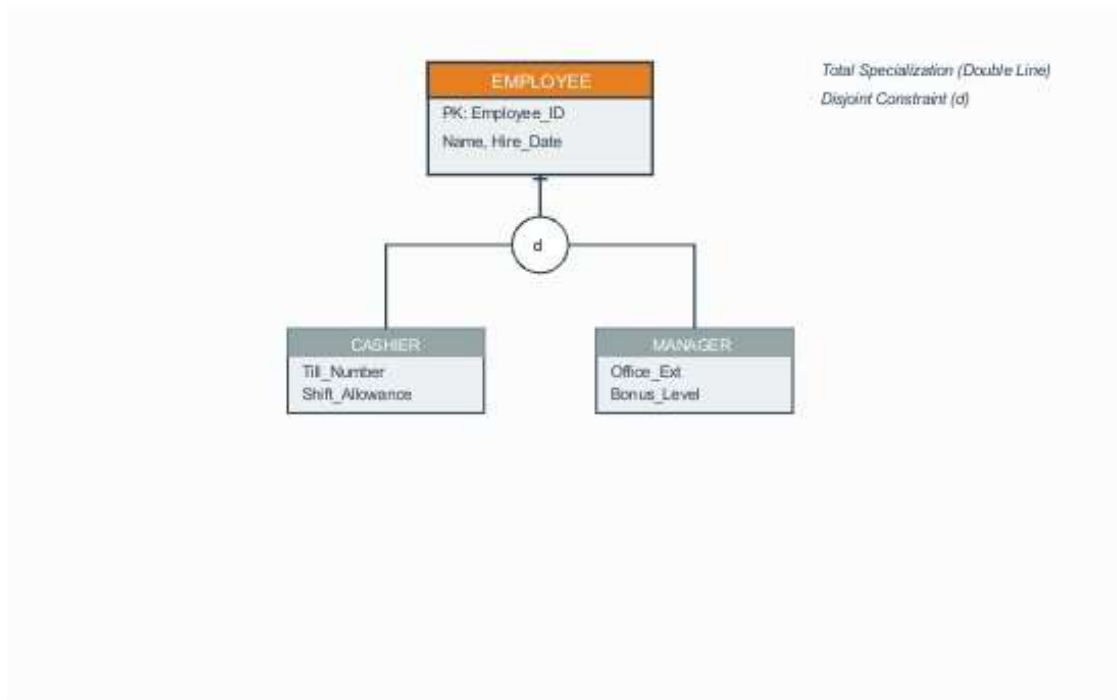


Figure 2. EER Diagram



Figure 3. Relations Between Tables